

VBcores IO Board v0.6

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VIN: 12-50 V DC
 MCU module: VB32G4 (STM32G474RE)
 2 × SPDT relay outputs, NC/COM/NO
 2 × high-side switched VIN outputs
 4 × 12 V stack-light outputs
 4 × buffered digital inputs
 2 × buffered analog inputs, 0-3.3 V
 I2C4 with selectable 3.3/5 V supply and pull-ups
 2 × parallel CAN/CAN FD connectors
 Input voltage monitoring, divider ratio 1:16
 Dimensions: 56 × 101 mm
 Mounting holes: 50 × 95 mm, D2.5 mm

12V, 0.5A/CH MAX OUT
 XH2.54 connector
 Designed for activity indicator tower with buzzer

Analog in 2ch, 3.3V MAX
 XH2.54 connector

Digital in 4ch, pulled-up
 XH2.54 connector

GPIO, 3.3V MAX
 Solder pads

Controlled power OUT
 (MOSFETs, OUT=VIN)
 XT30 compatible

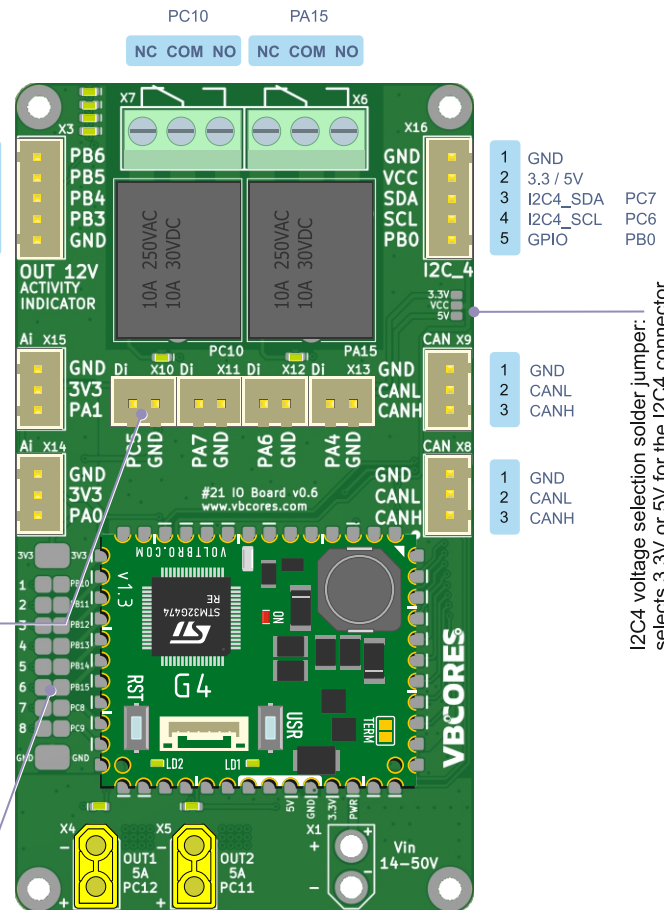
PC12 PC11

Power IN
 12-50V
 XT30 compatible

Input voltage sensing

Divider ratio: 1:16

Control	PIN	ADC
VIN_SENSE	PC0	ADC12_IN6



I2C4 voltage selection solder jumper:
 selects 3.3V or 5V for the I2C4 connector

NOTES:

1. X8 and X9 are wired in parallel and belong to the same CAN/CAN FD bus; they are not independent CAN interfaces.
2. Enable the optional 120 Ω TERM jumper only when the board is installed at a physical end of the CAN bus.
3. X4 and X5 are high-side switched VIN outputs. Their output voltage follows the board input voltage and is not regulated to 12 V.
4. X3 provides four switched 12 V source outputs. Maximum current is 0.5 A per channel; the total 12 V load, including the relay coils, is limited to 1.5 A.
5. Relay contact designations NC and NO refer to the de-energized state. Terminal order is NC-COM-NO.
6. Ratings printed on the relay case apply to the relay component; the allowable board-level load may be lower.
7. X10-X13 are active-low dry-contact inputs intended to be shorted to GND. Do not apply an external voltage to these inputs.
8. Maximum voltage on the X14 and X15 analog inputs is 3.3 V.
9. The X16 connector supply and I2C4 pull-up voltage are selected by the 3.3/5 V solder jumper. With the jumper open, connector VCC and pull-ups are not supplied.
10. PB0 on X16 is connected directly to the MCU and is not level-shifted. Use 3.3 V logic; its logic level does not follow the I2C4 jumper setting.
11. All interfaces and power outputs share board GND. Only the relay contacts are potential-free.
12. Relay, MOSFET, and 12 V control outputs are active HIGH and are held OFF during MCU reset by external pull-down resistors.



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